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Deciding on a College Transfer: Uncovering Transition Queries and Concerns via Reddit Topic Modeling

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ABSTRACT

Transitioning from community college to four-year university poses unique challenges for underrepresented STEM transfer students, such as credit transfer issues, adapting to new environments, and feeling out of place. Advisors often lack adequate resources, prompting students seeking online help. This study analyzed 628 academic subreddits on Reddit.com, resulting in a dataset of 664K posts, narrowed to 20K about college transfers. Topic modeling using three algorithms revealed broad and specific topics like finance, housing, lifestyle etc.

Sentiment analysis showed 10% negative posts, highlighting issues like difficult classes, class nearing capacity etc. These findings can aid advisors in better supporting transfer students.

KEYWORDS: Social media, Transfer students, Reddit, Sentiment Analysis, Topic Modeling

INTRODUCTION

Today, social media has become an indispensable part of our lives, particularly among students, who are generally heavy users. It has become a primary source of information and communication for most people — higher education students are no different. It has been identified that social media usage has both benefits and risks associated (Oye et al, 2012) for higher education students. Empirical studies have shown the major advantages of social media usage in higher education by enhancing communication, networking, and collaboration (Legaree, 2015). It has also created opportunities for rapid sharing of resources and provided an alternative platform for learning. However, regarding the purpose of using social media, David et al. (Oye et al, 2012) indicated that academic performance was adversely affected when social media sites were used for social and non-academic purposes only. Most of this line of work focuses more on the impact of social media on the academic performance of the students (Wang et al, 2011; Al-Rahmi & Othman, 2013; Lau, 2017). There are some works on identifying the social media influence on the transfer and career decisions of community college students (Davis et al, 2015). All the above-mentioned studies were heavily dependent on surveying or interviewing students to understand the impact of social media on their academic lives. No existing studies were found that analyzed social media data to understand the kinds of interaction and topics of conversation of students on social media, specifically for transfer students. This study taps into this gap to collect the first social media interaction dataset targeting specifically transfer students to understand their experience and challenges they face before and after transfer from their social media interactions.

Given the long-form forum format of the online discussion platform, Reddit, its “submissions” (posts) have been chosen as the primary target of our data collection. Reddit is also an ideal choice for its ability to provide users with pseudonymity, fostering open dialogue and authentic interactions across varied communities. Moreover, Reddit’s subreddit structure helps in finding relevant research data easily.

In this study, we first manually identified 628 academic subreddits which were then divided into five broad domains, (i) specifically related to college transfer, (ii) general college subreddits, (iii) a wide array of college-specific subreddits, (iv) some STEM major specific subreddits, and (v) general advising subreddits. Using Reddit’s PRAW API, a total of 664K posts were scraped from these subreddits. The posts were cleaned and filtered using zero-shot classification on Bidirectional and Auto-Regressive Transformers (BART)’s embedding of the dataset (Lewis et al, 2019) to separate 20K specifically college transfer-related posts covering 319 subreddits in total. A small subset (n = 100) of this dataset was manually labeled by three undergraduate and graduate research assistants which was used as a validation set for the parameter tuning. The final curated dataset is analyzed using topic modeling as it has the unique strength of analyzing educational textual data (Sakhovskiy et al, 2020), and it has been utilized to uncover latent topics from large scale data where other supervised machine learning methods fail because of the lack of labeled data. We employed and compared three topic modeling algorithms, such as unsupervised model Latent Dirichlet Allocation (LDA) (Blei et al, 2003), semi-supervised model

Correlation Explanation (CorEx) (Gallagher et al, 2017), and pre-trained model Bidirectional Encoder Representations from Transformers Topic (BERTopic) (Grootendorst, 2022) to identify the different factors (topics) that might influence the transfer decision and success of these students through the topic analysis of their queries and posts on Reddit. Additionally, sentiment analysis was employed using Valence Aware Dictionary and sEntiment Reasoner (VADER) (Hutto & Gilbert, 2014) to pinpoint the negative posts, aiding in identifying the key concerns students might have in the transfer process.

The main contributions are the following:

- Collecting and curating a social media dataset from Reddit specifically on college transfer.
- Identifying specific topics as well as broad topics related to college transfer to understand factors that influence students' transfer decisions and success using topic modeling.
- Analyzing the sentiments of the posts to pinpoint the key concerns of students in the transfer process.

The remainder of this paper is organized as follows, First, the LITERATURE REVIEW section discusses relevant literature and motivation behind this work. Following this, the METHODS section contains a discussion on the dataset preparation and methodology. Next, the RESULTS section summarizes and discusses results. Finally, the DISCUSSION AND CONCLUSION section provides a discussion of the findings, the significance and impact of this study, limitations and future work direction of the study, and concludes the paper.

LITERATURE REVIEW

Past research has majorly focused on the identification of factors that impact the successful transfer of underrepresented minorities in STEM and transfer students in general. Identified factors typically range over several domains such as interpersonal relationships, academics, and social support. However, much of this research relies on data acquisition from student surveys, interviews, and institutional data which provides quantitative measures of success such as GPA and successful graduation for factor identification (Blekic et al, 2020; Clausen & Wessel, 2015; Dika et al, 2015; Fauria & Fuller, 2015; Ishitani & Flood, 2018; Maliszewski & Hayes, 2020).

In (Townsend & Wilson, 2006), the authors identified that the transfer students initially require more assistance than they are given which impact their social integration and success in a new institution by conducting interviews with 19 students transferring from a small community college to a large research university. A similar study done by (Maliszewski & Hayes, 2020) explored the different sources of Transfer Student Capital (TSC) and how it was utilized by the students throughout their transfer process. This study collected data from 17 transfer students using interviews, eight observations of pre-transfer meetings between students and advisors, and a review of documents. They identified that the high schools, family members, and peers are influential sources of TSC, in addition to community college faculty and transfer advisors. In one study which chose to utilize survey data, the authors used the Laanan Transfer Students' Questionnaire (L-TSQ) to survey 280 STEM transfer students (Lopez & Jones, 2017). The survey consisted of several sections such as student background, experiences at their community college, and experiences at their current 4-year institution. The authors then conducted regression analysis on the dataset in order to find factors predictive of a students

academic adjustment and overall GPA. Their analysis found a correlation between factors such as parental educational level, faculty interaction, and transfer culture at the receiving university.

Existing literature mostly uses surveys and interviews as an instrument which uncovered that affordability, scholarship, socio-economic status, geographic location of the 4-year university, and GPA as the major factors (Wang et al, 2023) in transfer decision. The current methods provide significant insights into the transfer process, including determinants of transfer decisions, and post-transfer success factors. Nevertheless, there is a possibility that the survey instruments and interview questions may not encompass all crucial aspects of the problem while designing questions. Thus, engaging students in open-ended conversation within an informal environment could help identify additional information. Leveraging social media as a platform holds a promise to capture such interactions and also hear a greater number of voices than done in a survey or interview settings with a smaller population of transfer students.

Deng et al. (2022) emphasized the connection between social media usage in first-generation students and its corresponding impact on the student's academic experience. However, that study did not explicitly use social media data for their analysis but instead opted to use survey data from students. Analyzing social media data has been applied in the past to better understand public opinion, attitudes, and sentiment in several domains. This methodology was utilized in, (Melton et al, 2021) where the authors pulled data from 13 distinct subreddits related to the COVID-19 vaccine using the Reddit API. The data was then analyzed using sentiment analysis and latent Dirichlet Allocation (LDA) topic modeling to view the changes in relevant themes and opinions as they ranged over time and as a whole. Similarly, (Yan & Liu, 2022) conducted binary sentiment analysis on data collected from 8 subreddits that were connected to distinct universities to understand the effects of the COVID pandemic on the psychological well-being of individuals associated with higher education. Data was collected using the Pushshift API (now PullPush) and consisted of 165,570 total data points. This study applied model stacking to combine the prediction probabilities from the Robustly Optimized BERT approach (RoBERTa; Liu et al, 2019), and Graph attention networks (GAT; Veličković et al, 2017) to yield the final sentiments achieving an accuracy greater than 80%. While social media use and interaction have been studied as both a factor for student success and a possible resource for social support (Beck, 2016), to our best knowledge, there has yet to be a study that explicitly used social media data to identify factors that are important for student transfer decision and success. With this in mind, this paper offers a new direction in understanding these factors through the generation of a data set from social media, such as Reddit.com, and then conducting a corresponding analysis of the given factors by scraping hundreds of subreddit posts in the educational domain.

Motivation

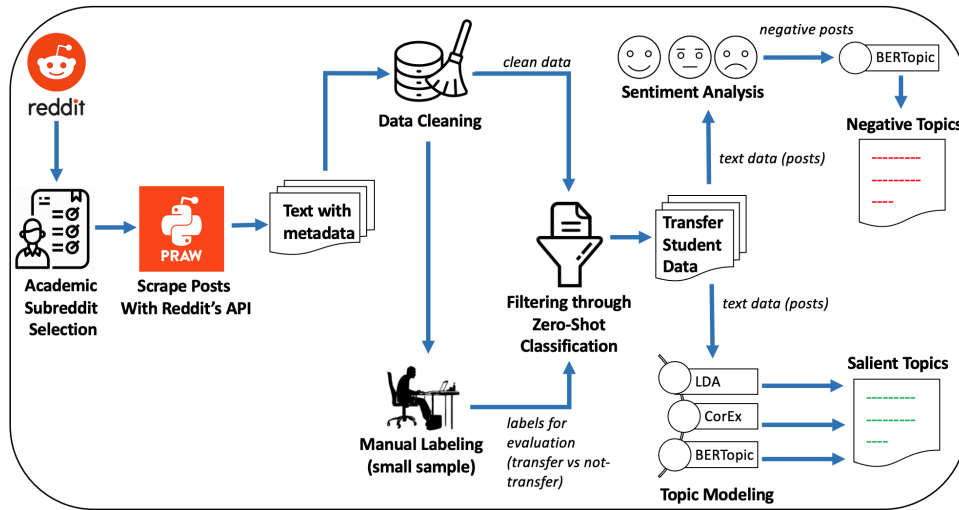
The motivation of this study is rooted in the fact that transfer students face unique challenges, specifically when applying to a four-year institution from a community college. Sometimes, their struggle persists even after transferring. They often need extra guidance on transferring credits, changing majors, adjusting to a new school, dealing with the fear of not fitting in, and many other issues. When searching for this information the most logical person to ask these questions is an advisor. However, advisors may not be fully equipped to answer some of those nuanced questions, and because of this, students tend to go to the internet to have these questions answered by people on social media. While the internet provides general guidance, personalized advice tailored to individual circumstances is scarce. The goal of this work is to

explore and analyze online posts to understand the kind of information transfer students are seeking on social media which affects their transfer decision and success. This information will help the transfer advisors, academic advisors, and student counselors to be better prepared and guide their students by providing more insight into their issues.

METHODS

The methodology used in this study consists of several phases, (i) data collection from the online discussion platform, Reddit, mostly from academic subreddits, (ii) data filtering to create a dataset focusing on transfer students, (iii) analyzing the sentiments of the Reddit posts, and (iv) unearthing the topics discussed in those posts. Figure 1 shows the overall workflow of the methods used.

Figure 1: Workflow - from data collection to generating results using Reddit data



Dataset

The dataset scraped for this study includes Reddit posts relating to and by students transferring from one college or university to another. These submissions contain questions, advice, rants, and useful information pertaining to transfer students. The dataset also contains other information from Reddit, such as the submission score, which is a ratio of upvotes and downvotes calculated by Reddit's algorithm.

Data Collection Methodology

This study was designed in Summer 2023. As a result of a site-wide overhaul of the Application Programming Interface (API) and its price rates as of July 5, 2023, the traditional data scraping methods for Reddit using third-party API wrappers became inaccessible during that period. Hence, Reddit's standard API interface, the Python Reddit API Wrapper (PRAW), was utilized instead for scraping which only provides access to 1000 posts per subreddit per API call. Since PRAW allows for only 1000 posts per subreddit, we expanded our selection of subreddits. In addition to this expansion, there are several API function calls to retrieve posts within PRAW. We sought to utilize the largest number of subreddits possible as well as the best few API calls

available. The following two sections describe the details of subreddit selection and API configurations.

Selection of Subreddits

There are some obvious target subreddits that contain pertinent data for analyzing the queries of transfer students, such as r/TransferStudents (13K+ subscribers), r/TransferToTop25 (10K+ subscribers), and r/CollegeTransfer (5K+ subscribers). Following closely behind are all the college subreddits, such as r/college, r/AskAcademia, r/ApplyingToCollege, and more (1,000K+ subscribers combined). While these subreddits are not specific to transfer students, they contain posts relating to college transfer. The next selection is the specific college subreddits where over 580 subreddits were chosen. These range from r/berkeley with over 134K subscribers to r/tesc with 1 subscriber. Many of the colleges represented by the 580+ specific subreddits have had a high number of transfer students in recent years. There are subreddits specific to college majors, for example, r/compsci and r/math which are geared toward their theoretical fields, but they do contain posts related to college transfer (2,000K+ subscribers together). Others, such as r/csMajors, and r/PhysicsStudents are involved in even more college transfer discussions and are quite large and active communities. Finally, there is a selection of general advice subreddits, such as r/NoStupidQuestions (4,000K subscribers), r/TooAfraidToAsk, r/Advice, and r/ask with active and large communities of users. While they generally don't involve transfer students, part of how the submission data is obtained from Reddit's API and data filtering steps involve filtering for the target domain, as our aim was to gather as many data points as possible related to transfer students. This list of subreddits is manually curated by an undergraduate research assistant.

API Calls

The next stage in data collection involved the Reddit API calls available through PRAW. Our implementation iterates over all the subreddits selected and performs the following API calls on each: (i) *hot*: returns the most recent and most liked posts in a subreddit, (ii) *top*: returns the most liked posts in a subreddit over a given timeframe. For the purposes of our research, the modifier 'all' was selected to indicate "all-time", (iii) *new*: returns the newest posts in a subreddit, and (iv) *search*: requires keywords or phrases to return posts containing them. This API call is limited to scrape data for the past 18 months. The keyword used in our implementation is "transfer".

After iterating over each API call for each subreddit, our implementation merges the lists of submissions while dropping duplicates. The result is anywhere from 0 to 4,000 posts from each subreddit with a portion containing the word "transfer", totaling 664,096 posts from 628 subreddits. The data is further classified and reduced to only contain posts relating to transfer students which is described in the next section.

Data Filtering through Classification

In the absence of enough hand-labeled data, Zero-Shot classification is utilized to filter the college transfer-related posts from the Reddit data. Zero-shot classification is a machine learning technique where a model is designed to classify data into classes that it has not seen during training. This is particularly useful when labeled data is scarce or when introducing new classes without the need for retraining the model. Specifically, the *facebook/bart-large-mnli*

(Lewis et al, 2019) model was employed for the task. BART (Bidirectional and Auto-Regressive Transformers) is a flexible model designed for a wide range of text generation and classification tasks. It was fine-tuned on the MNLI (MultiNLI) dataset for natural language inference tasks. The model extends conventional Transformer models by enabling both bidirectional and auto-regressive training. The full dataset scraped from Reddit is cleaned and classified using zero-shot classification on embedding generated by the BART model with the candidate labels, “transferring schools” vs “not transferring schools”.

Manual Labeling

A manual labeling approach is employed where three coders work in isolation to minimize bias and are exposed only to the post's body, cutting out any potential influence from other details like subreddit name, date, or labels from other coders. The manual labeling employs a binary table where posts about school transfers are marked as “Label Value” 1 and unrelated posts as “Label Value” 0. A small subset of 100 Reddit posts were labeled to evaluate the Zero-Shot classifier. These posts, identified by unique Reddit IDs, showed high confidence levels from the initial Zero-Shot classification model which is described in the RESULTS section.

Topic Modeling

Conventional approaches such as surveys and interviews, while valuable, may not capture all the topics involved in the decision-making process during the college transition period. Augmenting the established methods with the insights gained from social media data can potentially reveal previously unexplored topics. In this paper, the Reddit dataset is utilized to uncover topics in the college transition process. However, topic analysis is more subjective than sentiment analysis which poses some challenges. Hence, to extract topics from the data, three topic modeling algorithms are utilized and compared to detect interesting and specific topics. These algorithms range from an unsupervised LDA model to a semi-supervised CorEx model, and finally, BERTopic which is a pre-trained transformer model. The results from these models are aggregated and presented in Table 2.

LDA

The first model utilized is LDA (Blei et al, 2003) which operates by assuming that each document (Reddit posts) in the corpus is a mixture of various topics, and each topic is, in turn, a distribution of words. The algorithm iteratively assigns words in documents to topics based on their statistical distribution, with the goal of finding the most probable topic mix for each document and the word distribution for each topic. Words can be ranked by adjusting the model's alpha hyperparameter. The closer to 0 it is set, the more “unique” words will rank higher in the topic. This means that it will only display words that only appear in that specific topic and are absent or rare in other topics. This will allow that topic to be reviewed manually and determine if it is truly a new topic. An alpha of 0.75 was found to display relevant words while still maintaining strong coherent topics. LDA allows for custom ranges of topics to be produced from the corpus. The range used was 5 to 15. It was discovered that keeping the topics to be around 11-12 produced a good range of topics without being too redundant.

CorEx

The second algorithm used was CorEx (Gallagher et al, 2017) which aims to uncover latent topics in a corpus of text data while explicitly capturing and explaining the correlations between these topics. Unlike traditional statistical topic models like LDA, CorEx emphasizes the discovery of topics that are not only coherent but also semantically correlated. It does this by modeling the relationships between topics through a hierarchical structure, making it particularly useful for complex text datasets where topics are interconnected. It is also a semi-supervised algorithm. This means CorEx can be guided by a user to find hidden topics. Unsupervised models like LDA can sometimes overlook subtle topics due to them not being that prominent in the corpus. If the user has prior knowledge of the data, the algorithms can be guided to find more hidden topics. With CorEx you can provide anchor words that it will use as a basis to find topics that include those anchor terms. The initial anchor word used for the first analysis was “gpa”, then the list is extended to incorporate more anchor words derived from BERTopic as it is explained in the RESULTS Section.

BERTopic

The third model used was BERTopic (Grootendorst, 2022) which leverages the power of pre-trained BERT model embeddings to discover meaningful topics within a collection of text documents (Reddit posts). It operates in two main steps: document embedding and topic clustering. First, using a pre-trained BERT model, BERTopic converts each document into dense vector representations. This step captures the contextual information and semantic meaning of the words in the documents, making it effective for understanding the content. After obtaining the word embeddings of the document, BERTopic employs a clustering algorithm to group similar documents together into topics. These topics are shown as clusters of related documents. These clusters represent the discovered topics. Although BERTopic also provides a cluster of words like LDA, it derives those clusters contextually - like how humans would analyze topics. As a result of the nature of its analysis, BERTopic generates several topics automatically based on the entire collection of documents. The number of topics can be adjusted as a parameter and this variation was taken into consideration.

Sentiment Analysis

It is important to understand the sentiments of students turning to social media to divulge areas of concern. Researchers in the field of natural language processing have developed sentiment analysis models to determine the subjective sentiment of voluminous text data (Go et al, 2009). VADER is a rule-based sentiment analysis tool (Hutto & Gilbert, 2014). The VADER model returns scores which are floating point numbers that represent positive, neutral, and negative sentiments. In addition to individual scores, VADER also returns a compound score that covers all three sentiment categories in a single floating-point number between -1 and 1 where a negative score corresponds with negative sentiment and a positive score corresponds with positive sentiment. Our target here is to identify and analyze negative posts as manual inspection of a random selection of positive and neutral labeled posts was mostly found to be college transfer-related queries. We wanted to zoom into the negative posts to understand the adverse experiences of students during the transfer process.

RESULTS

Data Distribution

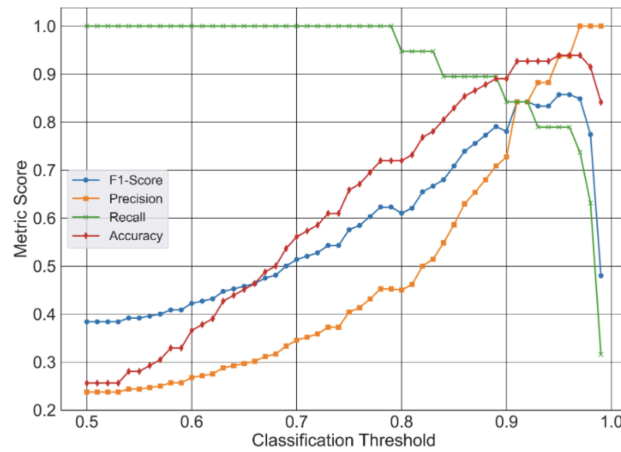
Upon completion of the data scrape across 628 subreddits, the data was analyzed for quality. Of the 628 initial subreddits, 622 had accessible submissions within the API calls utilized. The 622 subreddits were further classified into five categories: General Advice, General College, Specific College, Specific Major, and Transferring College. Table 1 provides a summary of the distribution of subreddits along with the number of posts in each of the categories.

Table 1: Distribution of subreddits and posts before and after filtering					
SUBREDDIT CATEGORY	NUMBER OF SUBREDDITS	NUMBER OF POSTS			
		BEFORE FILTERING		AFTER FILTERING	
		NUMBER	%	NUMBER	%
General Advice	5	10,541	1.59%	37	0.18%
General College	30	43,591	6.57%	934	4.51%
Specific College	556	563,049	84.81%	17,407	83.99%
Specific Major	27	39,284	5.92%	527	2.54%
Transferring College	4	7,431	1.12%	1821	8.79%

Results of Zero-shot Classification: Data Filtering

The analysis focuses on identifying students "transferring schools," with a confidence threshold of 0.5 the total count of posts is 156,392. However, a threshold of 0.95 was established through manual labeling and confusion matrix analysis to optimize evaluation metrics, such as precision, recall, F1 score, and accuracy. Figure 2 shows the measures with respect to confidence scores within the range [0.5,1]. At a 0.95 confidence level, we achieved the highest F1 score. With this threshold, 20,726 posts were identified as highly likely relating to college transfer. The model's performance is quite robust, yielding a precision of 0.94 and an F1 score of 0.86. However, the recall is somewhat lower at 0.79, indicating some false negatives where the model did not identify college transfer posts. However, for our study, we are more focused on filtering out false positives. Last two columns of Table 1 show the distribution of filtered posts from five categories of subreddits after applying zero-shot classification.

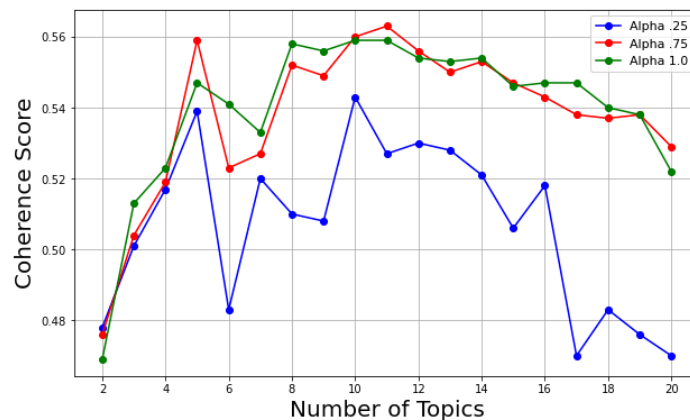
Figure 2: Performance metrics (precision, recall, F1 score, accuracy) across confidence threshold for zero-shot classification



Results of Topic Analysis

This section explains the findings from three topic modeling algorithms utilized in the study to identify queries and concerns related to college transfer. Out of the three topic modeling algorithms, LDA required the most amount of human intervention as it is necessary to filter enough general words, other than just the stop words to dig out meaningful topics. We tuned the hyperparameter, alpha, by evaluating the coherence score of topics against the number of topics for LDA. At 10-12 topics and higher alpha (0.75+), the coherence score reached above 0.5. Coherence score versus the number of topics for various alpha hyperparameters can be seen in Figure 3. After examining this plot, we picked alpha to be 0.75 and the number of topics to be 11 at a coherence score of 0.57 for LDA. However, upon review of the topics produced, LDA topic words were not conducive to contextual concepts. A lot of topics were incoherent having unrelated topic words using LDA, for example, one topic had its most relevant terms (film, mental, family, and money). Multiple other topics followed that trend.

Figure 3: LDA Coherence Scores vs. Number of Topics across various Alpha Hyperparameters



CorEx and BERTopic performed more coherently in finding relevant topics. BERTopic had a slight edge in being able to find a wider array of specific topics like tech recommendations, discussion about an online community (specifically on Discord), maintaining a healthy lifestyle,

health insurance, etc. Initially, with only 'gpa' as the anchor word, the semi-supervised model CorEx found more general topics such as financial situation, GPA requirements, social life, etc., that BERT was also able to find. We further improved CorEx with more vectors of anchor words selected from the results generated by BERTopic, such as ['aid', 'financial', 'scholarship', 'tuition'], ['group', 'discord', 'chat', 'server'], ['gym', 'fitness', 'climbing', 'workout'], ['friends', 'clubs', 'social', 'love', 'life', 'hang'], ['housing', 'campus', 'dorm', 'live', 'lease'], ['gpa', 'minimum', 'average'], ['laptop', 'ipad', 'tablet', 'notes'], ['insurance', 'health', 'coverage'], and ['chem', 'bio', 'physics', 'med', 'premed']. These helped in detecting some specific topics not identified by even BERTopic, such as mental health, career opportunities, communication, and research opportunities. All three topic modeling algorithms found similar broad topics such as finance, living on campus, and social life. Table 2 depicts aggregated topics across the three topic modeling algorithms with the union of topic words (most relevant ones are selected manually out of the top 10) if the topic is detected by more than one algorithm. The checkmark indicates the algorithm on that specific column was able to detect the corresponding topic.

Table 2: Aggregated (union) topics across three topic modeling algorithms with topic words				
TOPIC TITLE	TOPIC WORDS	LDA	CorEx	BERT
Social Life	friends, people, clubs, meet, party, social, love, life, hang	√	√	√
Living Arrangements	housing, campus, living, dorm, roommate, apartment	√	√	√
Financial Situation	aid, financial, scholarship, tuition, money, cost, save	√	√	√
Research Opportunities	research, opportunities, experience, career, program, advice		√	
Health Insurance	insurance, health, medicaid, coverage			√
Online Community	group, discord, chat, server, join			√
Tech Reco.	laptop, ipad, use, tablet, notes		√	√
GPA Requirements	average, minimum, high, gpa		√	√
Pre-med Questions	mcats, medical, med, premed, chem, bio, clinical, university		√	√
Picking a Major	program, major, degree, study, science, first, year	√		
Good Grades	stats, gpa, sat, act, grades, low, competitive		√	
Mental Health	mental, health, family, school		√	
Career Opportunities	career, opportunities, internships, major, job		√	
Communication	email, sent, received, weeks, ago, told, emailed		√	
Credit requirements	classes, credit, courses, taking, course, requirements, transferable	√	√	
Healthy Lifestyle	gym, fitness, climbing, workout			√
Switching Majors	major, switch, tech, internship, cs, business	√		
Online Class	take, class, online, time, plan	√		
Attending a State School	state, attend, pay, worth, tuition	√		

BERTopic was also utilized in an analysis of the negative sentiment Reddit posts produced by VADER. Although the posts labeled by VADER as negative sentiments (compound score

≤ -0.33) represented 10.32% ($n = 2,065$) of the filtered data, the topics within these posts have the largest material impact to students. Unique topics emerged specific to the negative sentiment data subset such as COVID-19, Parking & Amenities, Rejection (from admissions), Difficult Classes, and Class Nearing Capacity. Two topics emerged in the negative sentiment data that have some semantic overlap with the positive/neutral sentiment space: Enrollment & Orientation and Admissions Process. Upon manual review of the data assigned to these topics, the negative sentiment data subset contained distinctly negative posts about these topics including anxiety over the requirement to attend orientation prior to registration of classes when other students can secure their place earlier and confusion over ambiguous deadlines in the application process. Furthermore, there are also repeat topics such as GPA Requirements, Social Life, Living Arrangements, and Financial Situation. While these four topics are represented in the dataset at large, they also had strong representation among negative posts. The emergent topics in the negative sentiment subset point to areas in need of improvement across institutions.

Moreover, a manual inspection is performed into the negative posts pertaining to certain topics related to credit requirement and financial aid to understand why students are going to social media rather than going to advisors and what can be the impact of them seeking information there. From one post we identified that the student was not satisfied with the advice they received during the transfer process from the transfer advisor and hence seeking information in social media. However, the responses they receive in Reddit from multiple commenters were mixed and hence can create further confusion for the student. In another post a student was asking for specific information about FAFSA and the reply they received was misleading. These findings are important for the transfer student advisors and counselors to improve their advising and also for the higher education institutions to equip their advising staff with the necessary resources.

DISCUSSION AND CONCLUSION

This paper brings a new direction that uses social media data to identify factors that contribute to college transfer-related decisions. Traditional methods such as surveys and interviews may only uncover well known broad factors such as financial, academic, social, environmental, family education level etc. due to the limited scope of questioning. Also, most of such studies are performed on a smaller population of transfer students. Social media data analysis complements these traditional approaches, enhancing the identification of crucial elements in the college transfer process and also hearing the greater number of voices online than a traditional setting. Bringing in the social media discussions e.g. Reddit interaction in this paper, helps discover additional factors that contribute to the transfer decision and process. With the help of topic modeling on a curated dataset from Reddit, this paper identifies that students base their decision to transfer or have questions about living arrangements, research opportunities, career opportunities, switching majors, health insurance, mental health, tech recommendations, online communities (over discord) etc. Additionally, this paper conducts sentiment analysis to identify the negative Reddit posts to pinpoint major concerns students have while considering a transfer. Some of these concerns include dealing with ambiguous deadlines, outright rejection with little cause, difficult class requirements, and nonsensical procedures like mandatory orientation. Many of these subtle factors were not previously identified by existing research on transfer students using surveys and interviews. By uncovering these topics using social media data, this paper lays a foundational work to help counseling and advising personnel in the college transfer domain. The advisors and other administrators in higher education may be

interested to know about the queries that the students post on social media because of ineffective advising and how they can be misguided and become more confused with the responses they receive on social media.

The insights gained from this research will empower universities to refine their informational resources, directly addressing student concerns. Additionally, the identified topics will serve as a valuable resource for 2-year and 4-year university counselors, enabling them to better prepare and address students' queries regarding the transfer process. Finally, the results of this research can be utilized to design an automated advising system specifically for transfer students with the aim to deploy a multifaceted support system developed with cutting-edge AI and data analytics techniques tailored to the unique goals, circumstances, and academic history of students. This research work marks a significant step towards assisting transfer students on their academic journey. Upon request, we can share the Python scripts for curating the dataset used in this paper.

Limitations and Future Work

In April 2023, when this research team started collecting the data, Reddit announced its intention to add fees for its API access. Access to the Reddit API has been free since 2008. The announcement and subsequent enacting of the change led to site-wide protests and blackouts (Grantham-Philips, 2023). Due to this API change, several third-party applications were forced to cease activity, citing the exorbitant pricing. The third-party application, Pushshift (now PullPush), added the utility of retrieval of Reddit posts from specific subreddits covering an unlimited time frame (Baumgartner, 2020). However, because of the restriction of the capabilities of Pushshift by Reddit, it was a challenge to scrape the data from niche subreddits related to college transfers. An additional limitation is that it was not possible to identify posts only related to community college to four-year transfer vs. four-year to four-year transfer as we were not able to scrape all data from a niche *r/CollegeTransfer* and *r/TransferStudents* subreddit due to the Reddit's inhouse PRAW API limits.

In the future, we intend to identify the impact of the responses students are getting in their posts on Reddit. To identify that we need to analyze the response on some most up-voted as well as down-voted posts on Reddit related to college transfer. As the third party PullPush API is currently open to use it is now possible to collect more specific data related to college transfer from niche subreddits, such as *r/CollegeTransfer* and *r/TransferStudents*. This analysis of Reddit data will provide more insight into students' transfer decisions and their success by collectively recognizing the predominant environmental factors. As an important information source, social media has a substantial influence on younger generation students, and it is worth our investigation which will work as an environmental condition analysis for Social Cognitive Theory through stigma, misinformation, and peer influence detection. Although the analysis of curated dataset provides meaningful insights, more studies are needed to understand how these findings can help improve overall advising systems. In the future, we intend to disseminate our findings to the advisors so that they understand the discussions and sentiments of students related to university experiences. By identifying common concerns or emerging trends, advisors can proactively address issues that may impact student satisfaction or well-being. We plan to conduct focus group interviews with advisors at various universities and seek their insights about 1) their understanding of the identified topics and factors, 2) how they could incorporate these insights and overall methodology in their advising plan, and 3) understand their insights of new topics students are discussing and are not part of their advising

discussion. These interviews will help us validate our findings and integrate such methodology in the existing advising systems of the universities.

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